יום ניצוץ!
הפרויקטים הכי מבריקים בפקולטה

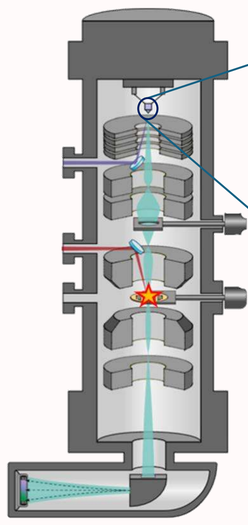
הפקולטה להנדסת חשמל ומחשבים ע"ש אנדרו וארנה ויטרבי

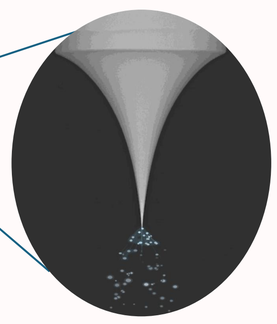
Passive Nanophotonic Doppler Cooling of Free Electrons

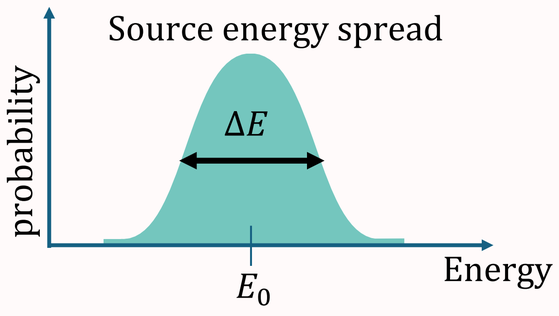
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The problem: Electron beam energy spread



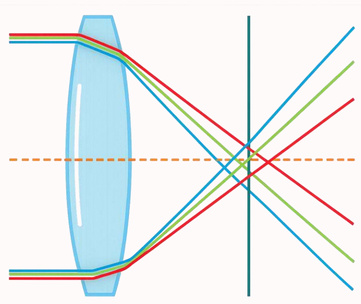




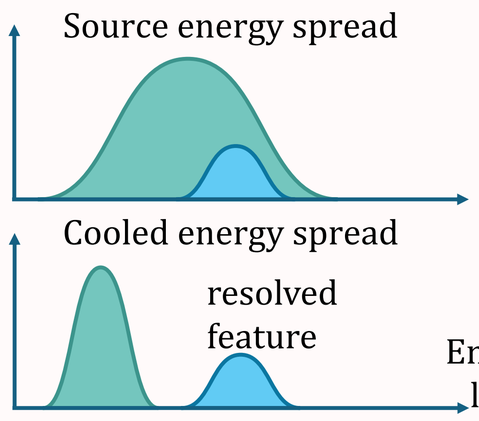
The electron beam introduces:
- **Limited flux** of electrons
- Initial electron energy uncertainty or **Energy spread**

Microscopy (imaging)
Energy spread causes **chromatic aberrations**



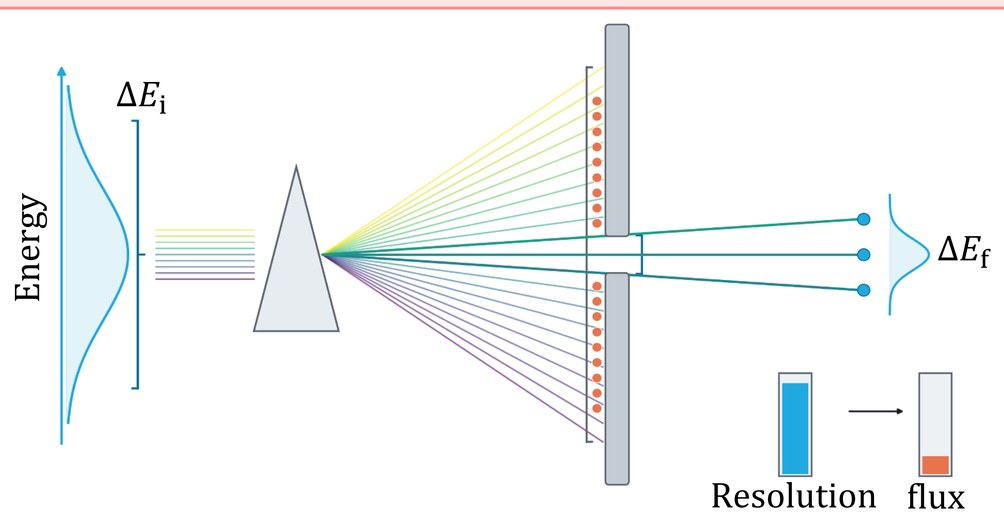


Spectroscopy (EELS)
Broad energy spread limit **spectral resolution** and hide weak features



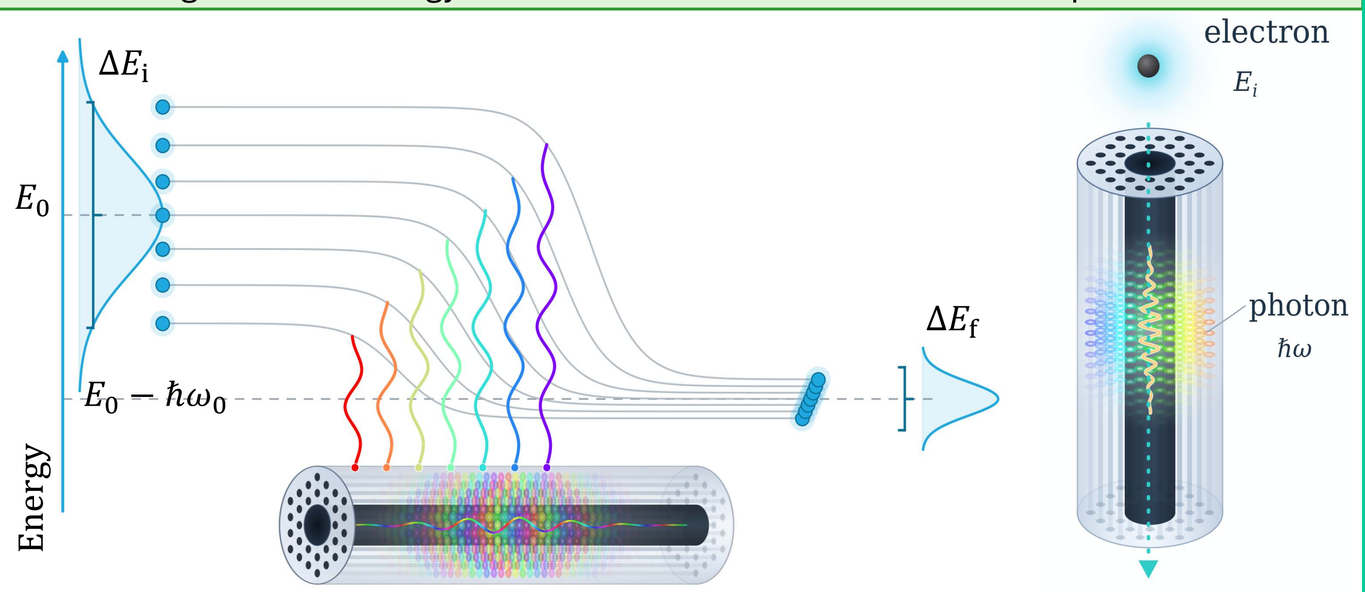
Limitations of classical monochromators

The resolution vs. flux trade-off
Monochromator filters electrons by energy



Our protocol: Optical monochromator mechanism

higher\lower energy electrons should emit blue\red shifted photons



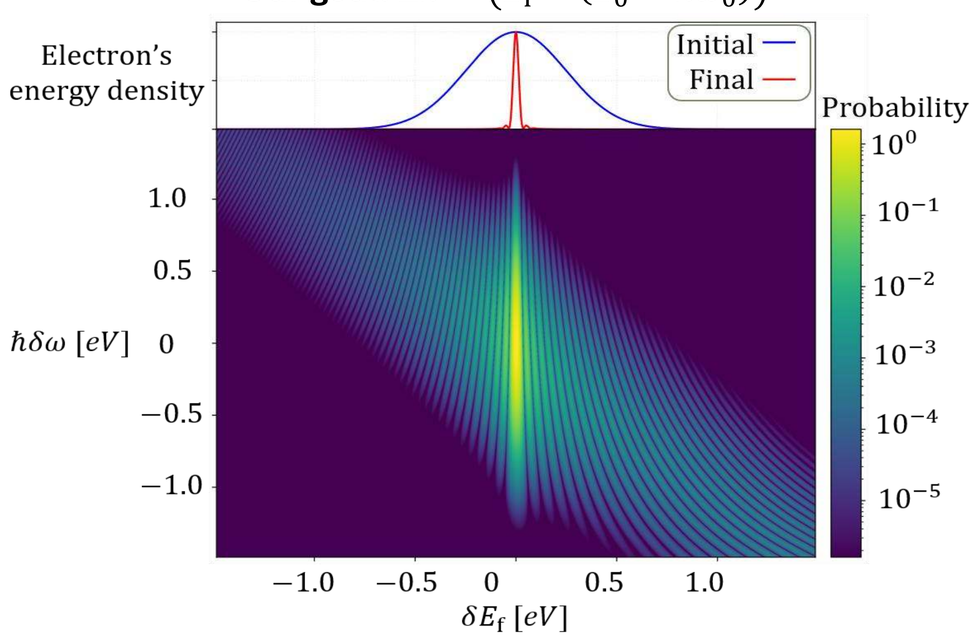
Solution by free electron quantum optics

Electron-photon interaction: $H_\mathrm{int} = \frac{e}{m}\hat{A}(\hat{\mathbf{r}}) \cdot \hat{\mathbf{p}}$

Spontaneous emission 1st order perturbation (weak coupling) transition amplitude:

- $M_{E_\mathrm{f},\omega} = -\frac{i}{\hbar}\int_{-\infty}^{\infty} dt\, e^{-i\left(\frac{E_\mathrm{i}-E_\mathrm{f}-\hbar\omega}{\hbar}\right)t}\langle E_\mathrm{f}, 1_\omega | H_\mathrm{int} | E_\mathrm{f}+\hbar\omega, 0_\omega\rangle$
  $\propto L_\mathrm{int}\mathrm{sinc}\left(\Delta k \frac{L_\mathrm{int}}{2}\right)$
- $\Delta k = k(E_\mathrm{f}+\hbar\omega) - k(E_\mathrm{f}) - q(\omega)$

Our goal: $\Delta k \propto \left(E_\mathrm{f} - (E_0 - \hbar\omega_0)\right)$

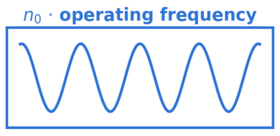


Resulting Cooling ratio : $\frac{\Delta E_\mathrm{f}}{\Delta E_\mathrm{i}} \propto \frac{L_0}{L_\mathrm{int}}$, $L_0 \propto \frac{2v_0 E_0}{\omega_0 \Delta E_\mathrm{i}}$

Dispersion engineering and design requirements

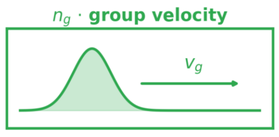
$$n_0 = \frac{c}{\omega_0}k(E_0) - \frac{c}{\omega_0}k(E_0 - \hbar\omega_0)$$

n_0 · operating frequency



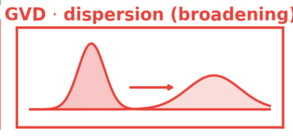
$$n_\mathrm{g} = \frac{c}{\hbar k'(E_0)}$$

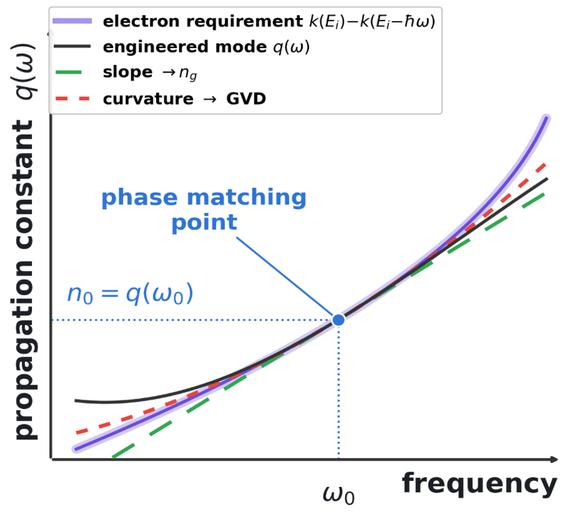
n_g · group velocity

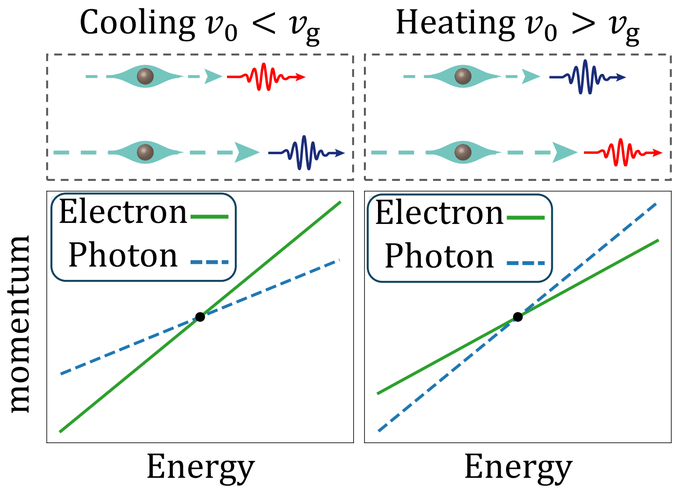


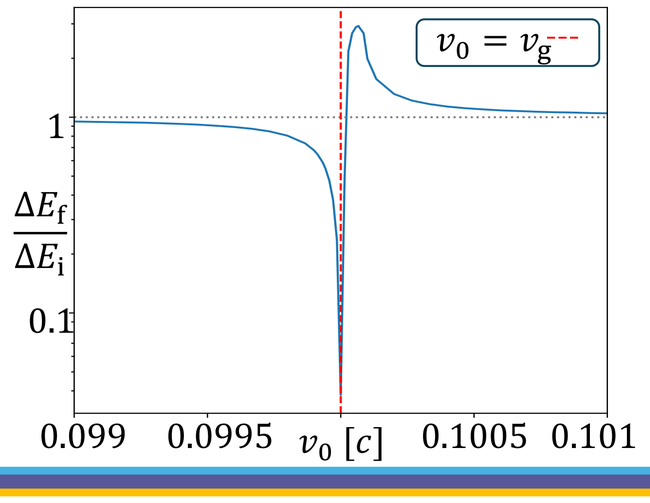
$$GVD = -\frac{\omega_0^2}{2\pi c}\hbar^2 k''(E_0)$$

GVD · dispersion (broadening)





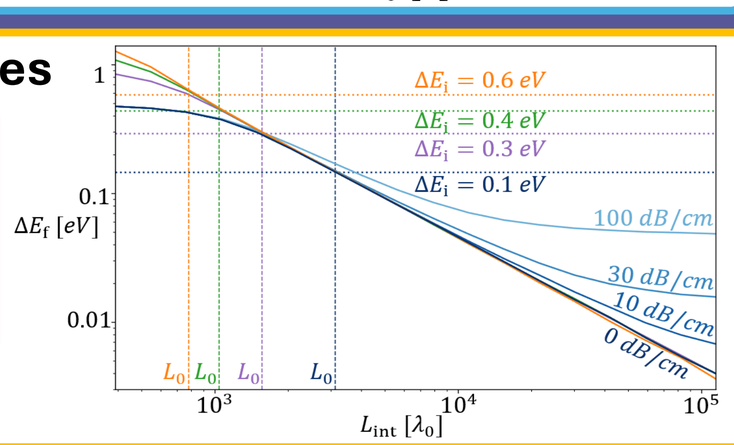




Limitation: photonic losses

Losses introduce saturation
- Losses limit the cooling ratio
- Saturation in the final width:

$$\Delta E_\mathrm{sat} = \frac{2E_0\alpha_\ell}{\omega_0}$$



Conclusions:

1. Free electrons can be cooled by spontanouse emission, an all-optical **analogue of atomic Doppler cooling**.
2. No inherent electron flux – resolution trade-off.
3. Cooling is set by the interaction length of the dispersion engineered waveguide, and is limited by photonic losses.
4. The photons inherit the electrons initial energy uncertainty.

References: [1] Hänsch and Schawlow, Opt. Commun. 13, 68 (1975). [2] Ruimy et al., Nat. Phys. 21, 193 (2025). [3] Karnieli et al., ACS Photonics 11, 3401 (2024). [4] Krivanek et al., Nature 514, 209 (2014). [5] Barwick et al., Nature 462, 902 (2009). [6] Konečná et al., Sci. Adv. 8, eabo7853 (2022).
[7] Shiran Even-Haim†, **Dor Gelem**†, Aviv Karnieli†, Renwen Yu, Ido Kaminer, Vladan Vuletić, and Shai Tsesses, in preparation.